Python and R Programming Rapid Fire Sheet

1. What is Python? Name some features of Python and explain how it is different than Java or any other languages?
2. What are the supported data types in Python? How to print datatype of a variable?
3. What is difference between list, tuple and dictionary? Is array in C is same as list in Python?
4. What is numpy and how it is better than list in python? What are ways of creating 1D, 2D and 3D arrays in numpy?
5. Write a code to convert List into Tuple in Python and reverse?
6. How can you get list of unique elements from a given list? Write a program.
7. What do you know about Flask & Django? How to implement a simple flask application?
8. How to connect MySQL from Python? Write code for executing DML and DQL queries.
9. What is exception? How to raise and handle exceptions in Python?
10. How to implement abstraction & encapsulation in Python?
11. Can we have public, private, protected, static, or final in Python classes?
12. How to read/write files in Python? Write code using 'with' statement.
13. Write a one-liner that will count the number of capital letters in a file.
14. Implement a simple server-client application using Python sockets.
15. Input a directory path from the user. List all text (.txt) files from that directory.
16. What is decorator? Where you have used it?
17. What is closure in Python?
18. What are high order functions?
19. Which different python packages you have used? How to install specific version of python package?
20. Write a program to print square roots of all even numbers in 1 to 100 using list comprehension. Write same code using lambda expressions.
21. Consider you have two arrays arr1 = (1,2,3) and arr2 = (2,3,4). Display pair of numbers having sum = 5.
22. Write a code to check if given string is palindrome or not.
23. Find number of occurrences of each character in the string (irrespective of its case).
24. Write a program to print terms of a Fibonacci series. How can you do the same using recursion?
25. What is pandas? Can you traverse through dataframes?

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26. How to convert a dataframe into CSV and Excel file? Which packages you need to use?
27. How will you create a new column in dataframe that have addition of two other columns in the dataframe? Suppose you have first_name and last_name as two different columns and create a new column that contains full_name?
28. Name data visualization packages in Python. How to create Pie chart from given data?
29. What is web scraping? How it is done in Python? Can we scrape data using xpath/CSS?

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